Generation of stable cell lines using DNA transposases

DNA transposons are mobile genetic elements that integrate transgenes into chromosomal DNA via a cut-and-paste mechanism. They are mobilized by transient expression of a transposase, which excises the donor sequence and inserts it into target sites through strand transfer reactions—without causing double-strand breaks at the insertion site.

Different transposase systems have been developed to support stable transgene integration in mammalian cells. These systems vary in integration efficiency, sequence preference, and genomic stability.

The Sleeping Beauty (SB) system (Ivics et al., 1997), including hyperactive versions such as SB100X (Mátés et al., 2009), integrates at near-random sites but has relatively low transposition efficiency and a tendency for local hopping, that is, reintegration close to the donor locus.

The piggyBac (PB) system (Ding et al., 2005), by contrast, transposes far more efficiently—in HEK293 cells about fourteen times as efficiently as the hyperactive SB11 (Wu et al., 2006), although the evolved SB100X closes much of that gap—and shows only weak local hopping. It inserts at TTAA tetranucleotides with no wider consensus, favouring transcription units and the neighbourhood of transcription start sites (Wilson et al., 2007), which makes it well-suited for insertional mutagenesis.

Risk assessment

– Work with human-derived material or transgenic cell lines (BSL-2)

▷ Wear gloves, safety glasses, lab coat

☐ Dispose waste after inactivation as REGULATED MEDICAL WASTE



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Procedures

Transfection of host cells

☐ 6-well plate
☐ Transfection reagent, non-viral
☐ Transposase helper plasmid
☐ Transfer plasmid

(1.) One day before transfection, seed cells as for any chemical transfection SOP 0012: about 4.2×10^5 cells per well of a 6-well plate.

(2.) Prepare a mix of the transposase and transfer plasmids at a sub-stoichiometric ratio.

Plasmid		Per well	
Transposase plasmid	Super PiggyBac transposase (SBI)	0.05 pmol	0.32 µg
Transfer plasmid	Custom	0.20 pmol	1.00 µg

Note: Transposase plasmids often encode a transiently expressed fluorescent protein to track transfection efficiency. The number of integrated copies, however, is set largely by the ratio of transfer to transposase plasmid, not by transfection efficiency.

(3.) Transfect the cells using a non-viral method such as Lipofectamine® SOP 0012.

(4.) After 6–12 h, replace the medium SOP 0012. 6–12 h

(5.) Begin selection or screening for stably transfected cells according to SOP 0019.

Hint: Begin selection 48–72 h after transfection, by which time transposition is complete. Sensitive lines benefit from the longer recovery.

Critical: The Limited Use Label License (LULL) of the piggyBac transposase (System Biosciences/Demeetra AgBio) applies to all derived cell lines and excludes any transfer to, use by, or contracted work for industry or commercial third parties.

Materials

Transfection of host cells

- ☐ **6-well plate**
- ☐ **Transfection reagent**, non-viral
  Hint: see SOP 0012 for selection of a transfection method.
- ☐ **Transposase helper plasmid**
  Hint: Super piggyBac is a hyperactive transposase variant (Yusa et al., 2011). The piggyBac transposase plasmid may not be redistributed; customers cannot propagate or re-clone the plasmid and must repurchase from System Biosciences (the academic distributor licensed by Demeetra AgBio) when stock is exhausted.
  - **Super PiggyBac Transposase Expression Vector**
    System Biosciences, PB210PA
- ☐ **Transfer plasmid**
  Hint: Addgene distributes piggyBac transfer plasmids under a Limited Use Label License (LULL) for academic non-profit research. No transfer to third parties, including institutional core facilities. The license runs until the earliest of the material being exhausted, Addgene losing the right to sell it, and the second anniversary of delivery, and sets out no renewal. Re-cloning the ITRs into a different backbone or synthesizing them de novo falls under Demeetra AgBio patents and requires a separate license.
  - *e.g.* **pPB-CAG-empty-pgk-hph**, constitutive CAG, hygromycin selection
    RRID Addgene_48753
  - *e.g.* **PB-CMV-MCS-Puro**, constitutive CMV, puromycin selection
    RRID Addgene_219794
  - *e.g.* **PB-TRE-EGFP-EF1a-rtTA**, Tet-On inducible (rtTA), puromycin selection
    RRID Addgene_104454

List of references

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Change log

2023-11-12 **Benjamin C. Buchmuller** Adaptation as SOP, synthesizing the Sleeping Beauty and piggyBac transposon literature cited in the body; donor-to-helper plasmid amounts from Shany Koren-Hauer (Tom Muir lab).

2026-10-26 **Benjamin C. Buchmuller** Credited Mátés et al. (2009), Wu et al. (2006), Wilson et al. (2007), Yusa et al. (2011) and the System Biosciences manual; corrected the name of the hyperactive Sleeping Beauty transposase, the local hopping and efficiency claims, the insertion-site preference, the seeding density, the medium change, the start of selection, and the name, scope and two-year term of the license.

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